

Belief Propagation Algorithm

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Abstract

The belief propagation is an iterative algorithm to solve complex problems with a large number of variables that are interdependent. Such a system can be described as a Bayesian network, which is a graph where each node represents a variable. Since this algorithm can be applied on any graphical model, the list of practical applications is long. Prominent examples for areas where the belief propagation is used is in artificial intelligence and information theory. Furthermore, it can be used to model and analyze financial markets and to make predictions, e.g. for risks in different scenarios. In our work, we focus on a problem from statistical physics. The nodes of the graph represent atoms in a crystal where the connection of the nodes represents the neighborhood of each atom. We calculate the spectral density and validate that our numerical result coincides with the analytical result.

1 Introduction

This project shows an application of the belief propagation algorithm in statistical physics. We derive the spectral density of a RRG via the belief propagation and compare it with the result from the direct diagonalization.

2 Theoretical Background

The cavity method corresponds to the belief propagation in statistical physics. In general, marginals are interdependent and can be represented by a system of nodes which are connected. The connectivity c specifies the number of nodes each node is connected with. A graph where every node is connected with c other nodes is called a random regular graph (RRG). Transferred to a physical system, each node represents one atom of a crystal which has c neighbors. In the cavity method we estimate the influence of the removal of one node. The node k with the node j removed is denoted as $\omega_k^{(j)}$ and called a cavity precision. We can use cavity precisions to derive the marginal distribution at node j . The Hamiltonian for our system reads:

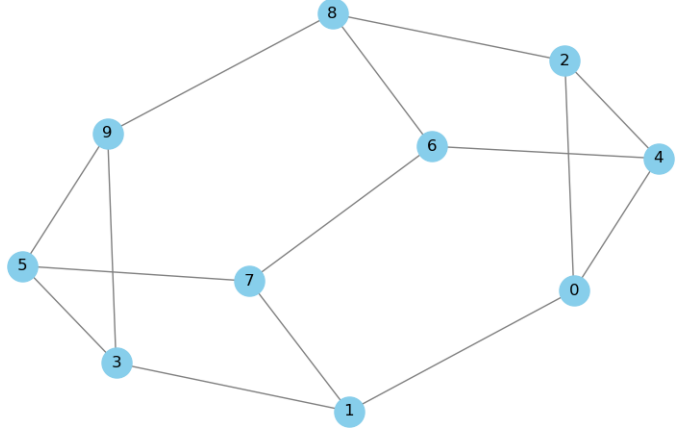


Figure 1: Graph of $N = 10$ nodes with connectivity $c = 3$

$$H = \sum_{i=1}^N E_i c_i^\dagger c_i - \sum_{\langle i,j \rangle} (c_i^\dagger c_j + c_j^\dagger c_i) = \text{diag}(E_1, \dots, E_N) - A, \quad (1)$$

where the energy depths of each atom are on the diagonal and the off-diagonal elements are determined by the adjacency matrix of the RRG. The latter represents the connection of the nodes, where an entry is one if two nodes are connected and zero if they are not. Note that we have no loops, i.e. there is no node which has a connection to itself. Based on this Hamiltonian, we can write a recursive equation for each cavity precision. Once equilibrium is reached, the marginals ω_j can be calculated based on the cavity precisions:

$$\boxed{\omega_k^{(j)} = i(\lambda - i\varepsilon - E_k) + \sum_{l \in \partial k \setminus j} \frac{1}{\omega_l^{(k)}}} \quad \boxed{\omega_j = i(\lambda - i\varepsilon - E_j) + \sum_{l \in \partial k} \frac{1}{\omega_l^{(j)}}} \quad (2)$$

with the parameters λ and ε . $\partial k \setminus j$ denotes the neighbouring nodes of node k except node j . From the marginals the spectral densities can be derived via:

$$\rho(\lambda) = \sum_{j=1}^N \frac{\text{Im } G_{jj}}{N \cdot \pi}, \quad (3)$$

with $G_{jj} = \frac{i}{\omega_j}$.

3 Implementation and Results

The implementation was made in Python 3.11 with the environment 'Spyder'. We designed a function that constructs a RRG via the 'networkx' module and turns the graph into the adjacency matrix. Adding the energy depths on the diagonal yields the hamiltonian Eq. (1). Another function gets the indexes of the neighbors for each node, which are used in another function that runs the recursive equation for the cavity precisions, as given in the LHS of Eq. (2). For numerical stability we introduce a parameter $\alpha \in [0,1]$ which weights the old result at $t-1$ and the new result at t via $(1-\alpha)\omega_k^{(j)}(t-1) + \alpha\omega_k^{(j)}(t)$. We found that $\alpha = 0.2$ yields stable results without extending the equilibration time too much. The changes between the updated and the old cavity precisions are tracked for each iteration cycle. Once all cavity precisions have a difference below $\epsilon = 10^{-6}$, the iteration breaks and the marginals are calculated from the equilibrated cavity precisions according to the RHS of Eq. (2). Running this algorithm for various λ -values yields the spectral density.

We compare the result from the cavity method for $\epsilon = 10^{-3}$ with the exact result from the direct diagonalization of the hamiltonian matrix. The system has a size of $N = 2^{11}$ atoms/nodes. In Fig. 2 we can see that both results are in agreement with each other.

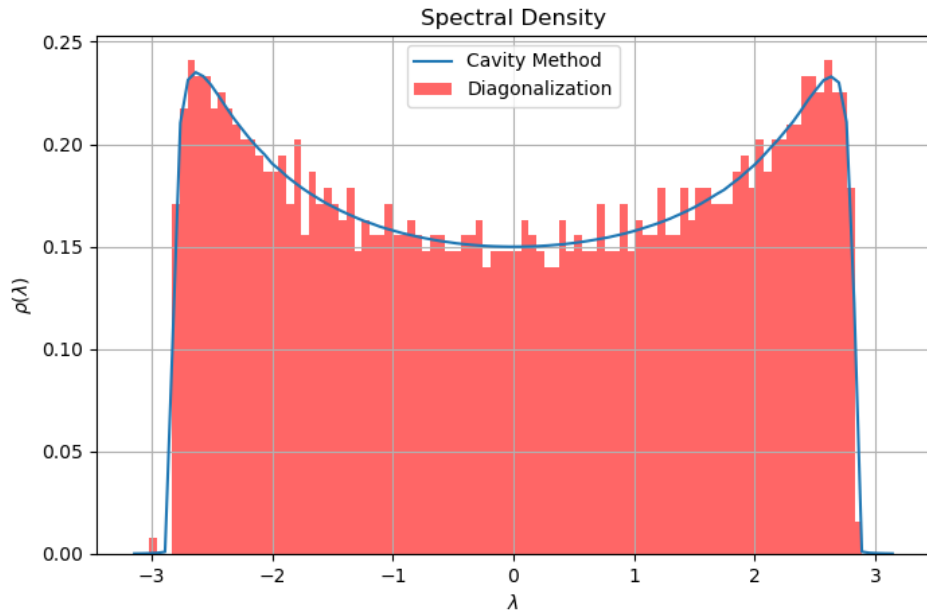


Figure 2: Comparison between the cavity method and the direct diagonalization.

4 Outlook

We will create various samples for different disorders to model the transition from an insulator to a conductor.